

Xv6 and Unix Utilities

Name: Sawera Iftikhar

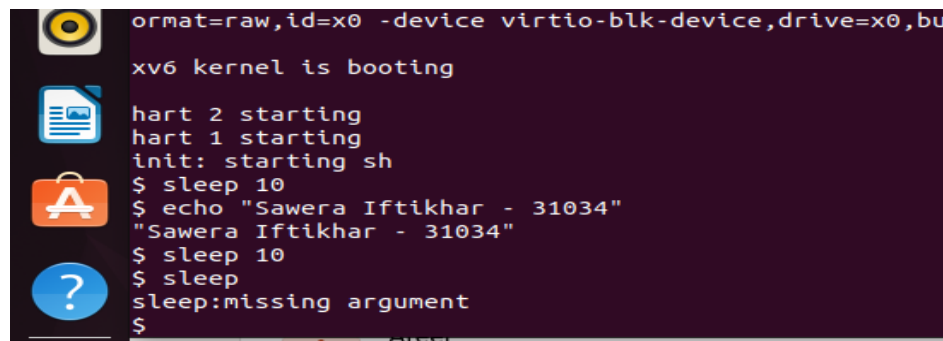
ERP: 31034

Repository: SaweraIftikhar/xv6-labs-2026

Problem Set #1

Branch: `problemSet1`

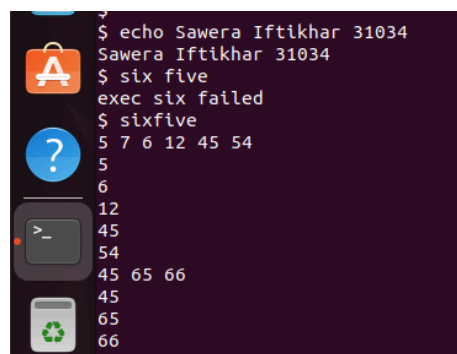
1. Sleep



```
format=raw,id=x0 -device virtio-blk-device,drive=x0,bu
xv6 kernel is booting
hart 2 starting
hart 1 starting
init: starting sh
$ sleep 10
$ echo "Sawera Iftikhar - 31034"
"Sawera Iftikhar - 31034"
$ sleep 10
$ sleep
sleep:missing argument
$
```

2. Sixfive

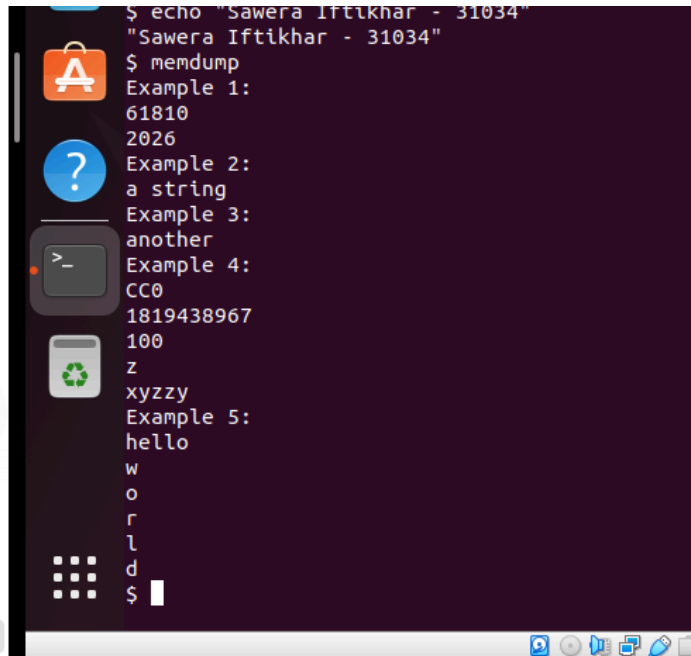
Screenshot of the `sixfive` program running in xv6.



```
$ echo Sawera Iftikhar 31034
Sawera Iftikhar 31034
$ six five
exec six failed
$ sixfive
5 7 6 12 45 54
5
6
12
45
54
45 65 66
45
65
66
```

3. Memdump

Screenshot of the `memdump` program running in xv6.



```
$ echo "Sawera Iftikhar - 31034"
"Sawera Iftikhar - 31034"
$ memdump
Example 1:
61810
2026
Example 2:
a string
Example 3:
another
Example 4:
CC0
1819438967
100
z
xyzyy
Example 5:
hello
w
o
r
l
d
$
```

4. Find

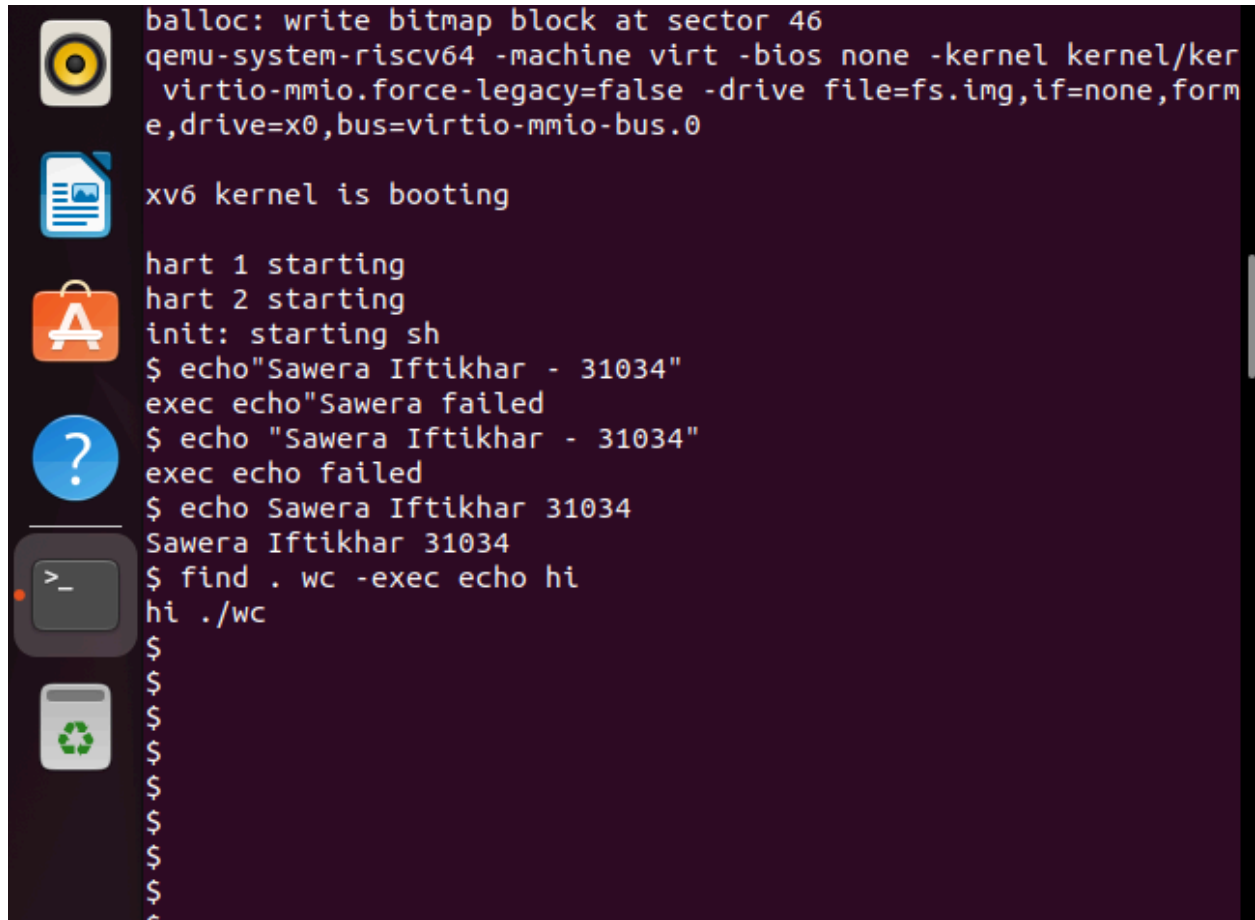
Screenshot of the `find` program running in xv6.



```
$ -nographtc -global virtio-mmio.force-legacy=raise -drive rtc
format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-
xv6 kernel is booting
hart 1 starting
hart 2 starting
init: starting sh
$ echo "Sawera Iftikhar - 31034"
"Sawera Iftikhar - 31034"
$ echo > b
$ mkdir a
$ echo > a/b
$ mkdir a/aa
$ echo > a/aa/b
$ find . b
./b
./a/b
./a/aa/b
$
```

5. Find with Exec

Screenshot of the `find -exec` extension running in xv6.



```
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/ker
virtio-mmio.force-legacy=false -drive file=fs.img,if=none,form
e,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

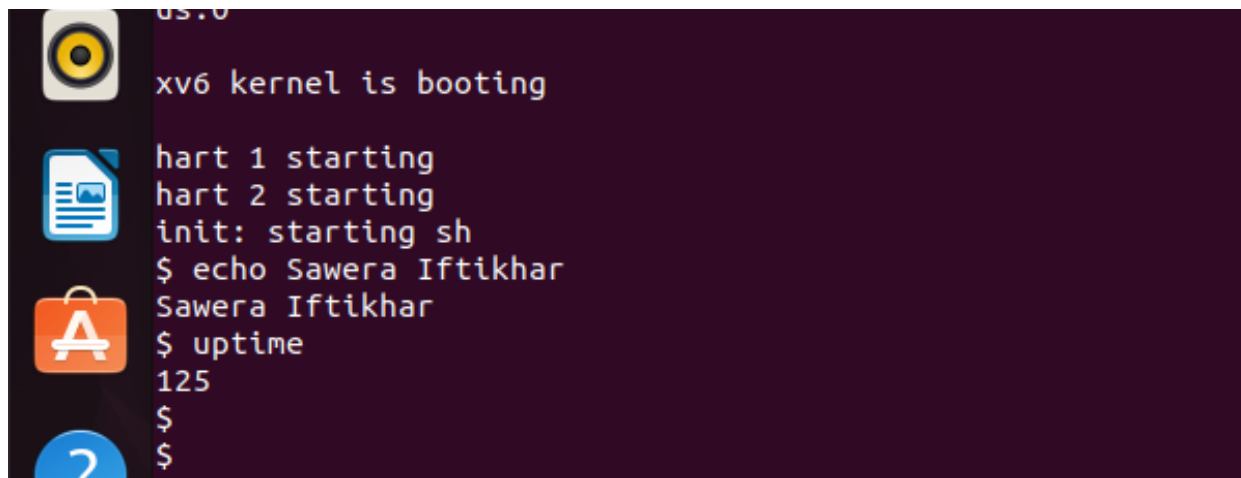
hart 1 starting
hart 2 starting
init: starting sh
$ echo "Sawera Iftikhar - 31034"
exec echo "Sawera failed"
$ echo "Sawera Iftikhar - 31034"
exec echo failed
$ echo Sawera Iftikhar 31034
Sawera Iftikhar 31034
$ find . wc -exec echo hi
hi ./wc
$
$
$
$
$
$
$
$
$
```

First Assignment

Branch: `assignment1`

1. Uptime

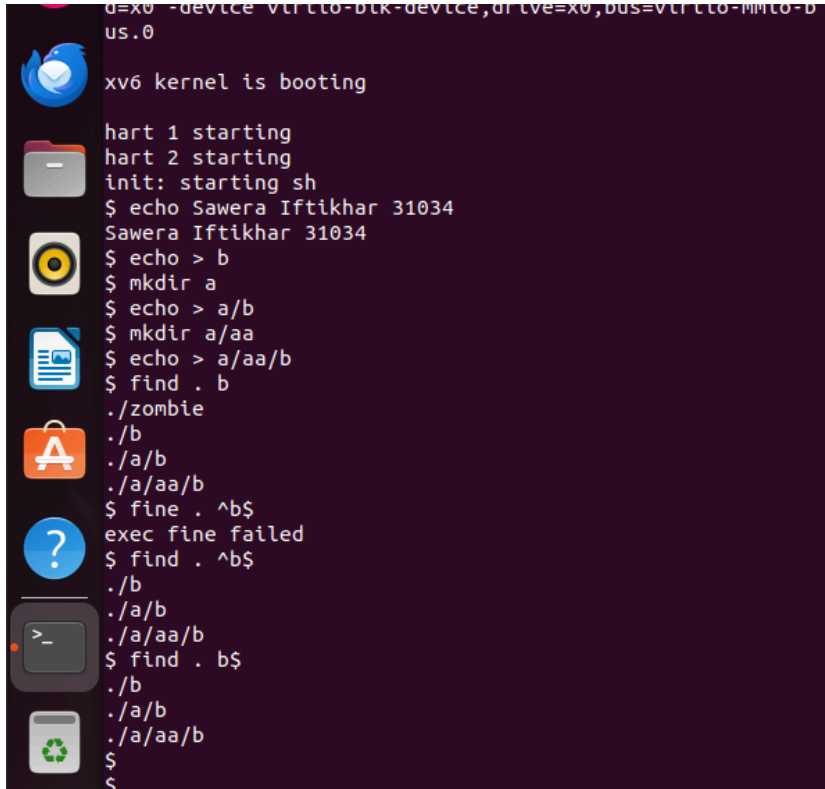
Screenshot of the `uptime` program running in xv6.

A screenshot of a terminal window with a dark purple background. On the left side, there is a vertical dock with four icons: a yellow and black target icon, a blue document icon, an orange shopping bag icon, and a blue circle with a white question mark icon. The terminal text is as follows:

```
OS.0
xv6 kernel is booting
hart 1 starting
hart 2 starting
init: starting sh
$ echo Sawera Iftikhar
Sawera Iftikhar
$ uptime
125
$
$
```

2. Regex Support in Find

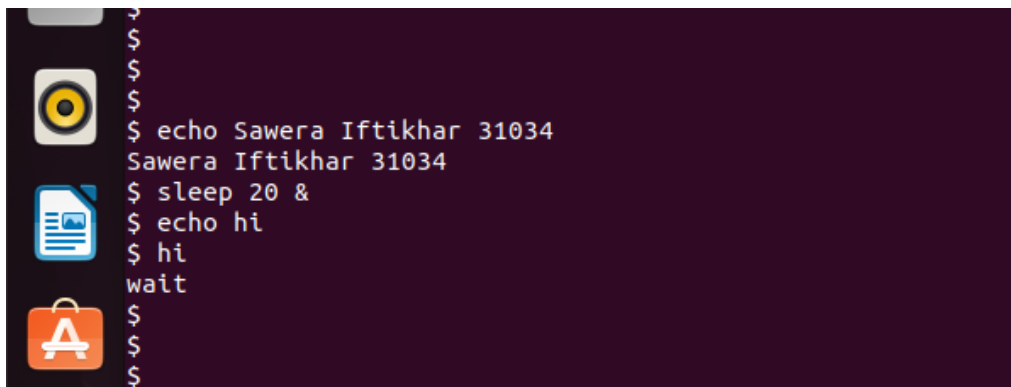
Screenshot of the modified `find` program supporting regular expressions.



```
d=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0
xv6 kernel is booting
hart 1 starting
hart 2 starting
init: starting sh
$ echo Sawera Iftikhar 31034
Sawera Iftikhar 31034
$ echo > b
$ mkdir a
$ echo > a/b
$ mkdir a/aa
$ echo > a/aa/b
$ find . b
./zombie
./b
./a/b
./a/aa/b
$ fine . ^b$
exec fine failed
$ find . ^b$
./b
./a/b
./a/aa/b
$ find . b$
./b
./a/b
./a/aa/b
$
$
```

3. Wait Support in Shell

Screenshot of the modified xv6 shell supporting the `wait` command.



```
$
$
$
$ echo Sawera Iftikhar 31034
Sawera Iftikhar 31034
$ sleep 20 &
$ echo hi
$ hi
wait
$
$
$
```